30/05/2025

Code-A_Phase-1

Aakash

Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***011-47623456

MM : 240 | TYM for NEET-2025-26_Practice Test-02A | **Time : 60 Min.**

Topics Covered:
Physics: Motion in a Straight Line: Differentiation, Application of Differentiation (Slope, Maxima & Minima), Integration, Application of Integration (Area under curve), Additional Mathematical Concepts, Graphical Section: Straight Line, Parabola, Ellipse, Circle, Hyperbola
Chemistry: Some Basic Concepts of Chemistry: Stoichiometry and Stoichiometric calculations, Calculations regarding limiting reagents, Reactions in solutions: Mass percentage or weight percentage, Mole-fraction, Molarity, Molality, Normality
Botany: Cell: The Unit of Life (Contd.): Cell Wall, Endomembrane system - endoplasmic reticulum, Golgi body, lysosome, vacuole, mitochondria, plastid, ribosome, cytoskeleton, centrosome and centrioles, cilia and flagella, Nucleus, Chromosome, Microbodies
Zoology: Structural Organisation in Animals: Animal tissues-II: Muscular tissue, Nervous tissue, Biomolecules-I: (Upto Polysaccharides)

General Instructions :
Instructions:
(i) Use blue/black ballpoint pen only to darken the appropriate circle.
(ii) Mark should be dark and should completely fill the circle.
(iii) Dark only one circle for each entry.
(iv) Dark the circle in the space provided only.
(v) Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on Answer sheet.
(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

1. The minimum value of $y = x^2 - 4x$ will exist at x is equal to
(1) 1
(2) 2
(3) $\frac{1}{4}$
(4) $\frac{1}{2}$

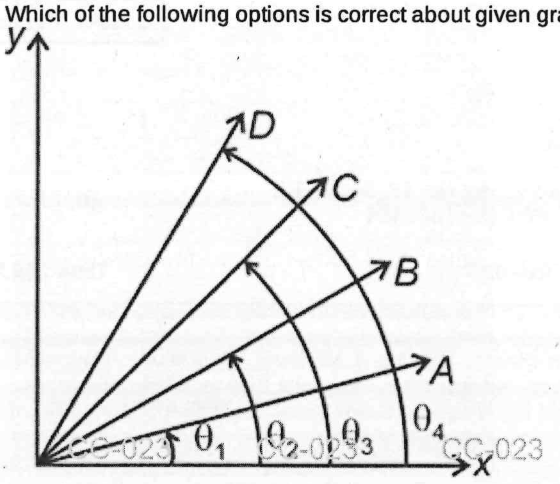
2. If $y = 3t^2 + 2t$, then value of $\int_0^2 y\,dt$ will be
(1) 4
(2) 8
(3) 2
(4) 12

3. If $y = \sin\theta$ and $x = \cos\theta$, then the value of $\frac{dy}{dx}$ at $\theta = \frac{\pi}{4}$ will be
(1) 1
(2) –1
(3) $\frac{1}{\sqrt{3}}$
(4) Zero

4. For the equation $y = x^3$ (where x and y are in m), the slope of the curve at $x = 1$ m will be
(1) 3
(2) 1
(3) $\frac{1}{3}$
(4) Zero

5. Area under the curve $y = 3x^2$ between the coordinates $x = 0$ to $x = 3$ m is
(1) 8 m^2
(2) 18 m^2
(3) 27 m^2
(4) 36 m^2

6. Which of the following options is correct about given graph?



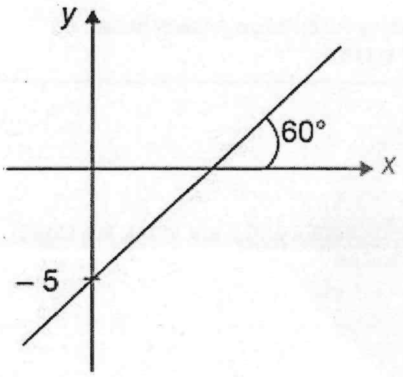
(1) *A* and *B* have minimum value of slope

(2) *B* has minimum value of slope

(3) *D* has maximum value of slope

(4) *A*, *B*, *C* and *D* have same value of slope

7. The graph of *y* vs *x* is given below. The value of slope and intercept respectively are



(1) $\sqrt{3}, -5$

(2) 5, $\sqrt{3}$

(3) $-\sqrt{3}, 5$

(4) $\sqrt{3}, 5$

8. Evaluation of $\int_0^3 t^2 dt = I$, then I will be equal to

(1) Zero

(2) 3

(3) 9

(4) 27

9. The maximum value of $(\sin\theta + \sqrt{3}\cos\theta)$ is

(1) 2

(2) 1/2

(3) $\sqrt{3}$

(4) $\frac{1}{\sqrt{3}}$

10. A : Differentiation can be used to find maxima or minima of a function.
R : At maxima or minima, differentiation of a function is negative.

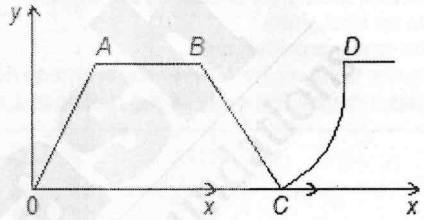
(1) If both Assertion & Reason are true and the reason is the correct explanation of the assertion, then mark **(1)**.

(2) If both Assertion & Reason are true but the reason is not the correct explanation of the assertion, then mark **(2)**.

(3) If Assertion is true statement but Reason is false, then mark (3).

(4) If both Assertion and Reason are false statements, then mark (4).

11. In the following graph, the slope of curve *AB*, *BC* and *CD* are



(1) Zero, positive, negative

(2) Zero, zero, variable

(3) Positive, negative, variable

(4) Zero, negative, variable

12. Differentiation of $\sqrt{x}$ w.r.t. *x* is

(1) x^2

(2) $\frac{\sqrt{x}}{2}$

(3) $\frac{1}{2x}$

(4) $\frac{1}{2\sqrt{x}}$

13. If $y = \sin 4x$, then value of $\frac{dy}{dx}$ will be

(1) $4\cos 4x$

(2) $-4\cos 4x$

(3) $\frac{\cos 4x}{4}$

(4) $-\frac{\cos 4x}{4}$

14. If $(x + 2)^2 + (y - 1)^2 = 9$ represent the equation of a circle, then coordinates of centre of circle is

(1) (1, 2)

(2) (–2, 1)

(3) (2, 1)

(4) (1, 1)

15. Sum of first n natural numbers is given by

(1) $\frac{n}{2}$

(2) $\frac{(n-1)}{2n^2}$

(3) $\frac{n(n+1)}{2}$

(4) $\frac{(n+1)}{2}$

CHEMISTRY

16. The mass of zinc required to produce 2.24 L of hydrogen gas at STP on treatment with excess of dilute HCl is (Atomic mass of Zn = 65.4 u)

(1) 13.0 g

(2) 6.54 g

(3) 3.25 g

(4) 19.54 g

17. Volume of oxygen required at STP to react completely with 160 g of methane gas according to following reaction is
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

(1) 224 L

(2) 44.8 L

(3) 22.4 L

(4) 448 L

18. If 4 g of NaOH is dissolved in 36 g of H_2O, then the mole fractions of NaOH and H_2O in the solution respectively are

(1) $\frac{2}{21}, \frac{19}{21}$

(2) $\frac{1}{20}, \frac{19}{20}$

(3) $\frac{2}{20}, \frac{18}{20}$

(4) $\frac{1}{21}, \frac{20}{21}$

19. 10 g of $H_2(g)$ and 64 g of $O_2(g)$ on complete reaction gives how much mass of water?

(1) 36 g

(2) 54 g

(3) 90 g

(4) 72 g

20. Concentration term which depends on temperature, is

(1) Molality

(2) Mole fraction

(3) % w/v

(4) % w/w

21. How many ml of 1 M H_2SO_4 is required to neutralise 20 ml of 1 M NaOH solution?

(1) 20 ml

(2) 2.5 ml

(3) 10 ml

(4) 5 ml

22. For the following reaction, $H_2(g) + Cl_2(g) \rightarrow 2\,HCl(g)$, if initially 20 mL of $H_2(g)$ and 20 mL of $Cl_2(g)$ are taken in the container, then the volume of the gas present in the container after completion of the reaction is

(1) 40 mL

(2) 10 mL

(3) 50 mL

(4) 15 mL

23. Mole fraction of urea in its aqueous solution containing 10% urea (w/w) is

(1) $\frac{5}{24}$

(2) $\frac{2}{15}$

(3) $\frac{1}{31}$

(4) $\frac{1}{20}$

24. 56 g of nitrogen reacts with 10 g of hydrogen to produce ammonia. Mass of ammonia formed is

(1) 68 g

(2) 34.8 g

(3) 56.7 g

(4) 66 g

25. 300 g of C_2H_6 is burnt in excess of oxygen. Mass of CO_2 produced is

(1) 110 g

(2) 220 g

(3) 440 g

(4) 880 g

26. If 30% (mass/mass) aqueous solution of KI has density 1.02 g/mL then molarity of the solution will be (Atomic mass of K = 39 u and I = 127 u)

(1) 2.8 M

(2) 1.4 M

(3) 2.5 M

(4) 1.8 M

27. What quantity of 80% pure $CaCO_3$ will give 28 g of CaO on heating?

(1) 50 g

(2) 75 g

(3) 62.5 g

(4) 56.2 g

28. Number of glucose molecules present in 200 mL of 0.4 M glucose solution is

(1) 4.8×10^{21}

(2) 4.8×10^{22}

(3) 1.8×10^{23}

(4) 1.8×10^{24}

29. In one molal solution that contains 0.5 mole of a solute, there is

(1) 100 mL of solvent

(2) 1000 g of solvent

(3) 500 mL of solvent

(4) 500 g of solvent

30. 12.04×10^{22} molecules of urea are present in 500 mL of its solution. The molarity of the solution is

(1) 4 M

(2) 0.4 M

(3) 2.5 M

(4) 0.2 M

BOTANY

31. Select the **incorrect** statement w.r.t. functions of cell wall.

(1) Protection from mechanical injury

(2) Cell to cell interaction

(3) Maintenance of the shape of cell

(4) Does not allow material to pass in and out of the cell

32. "The central part of centriole is made up of <u>A</u> and is called <u>B</u>."

Select the option which **correctly** fills the blanks

	A	B
(1)	Microtubule	Hub
(2)	Microtubule	Central sheath
(3)	Protein	Central sheath
(4)	Protein	Hub

(1) (1)

(2) (2)

(3) (3)

(4) (4)

33. How many doublets of microtubules are arranged peripherally in the axoneme of flagella?

(1) 18

(2) 22

(3) 20

(4) 9

34. Which among the following is the function of the cytoskeleton?

(1) Protein synthesis

(2) Motility

(3) Lipid synthesis

(4) mRNA synthesis

35. Nucleolus is the site of the synthesis of

(1) rRNA

(2) tRNA

(3) DNA

(4) mRNA

36. Carotenoids containing colored plastids are

(1) Elaioplasts

(2) Chromoplasts

(3) Aleuroplasts

(4) Amyloplasts

37. The mitochondrial matrix has all, except

(1) Circular DNA

(2) A few RNA molecules

(3) 80S ribosomes

(4) Enzymes for aerobic respiration

38. Tonoplast is the

(1) Single membrane around lysosome

(2) Double membrane complex around lysosome

(3) Single membrane around sap-vacuole

(4) Double membrane complex around sap-vacuole

39. Endomembrane system in eukaryotic cells include

(1) Single and double membranous cell organelles.

(2) Endoplasmic reticulum, Golgi complex, lysosome and vacuole.

(3) Non-coordinated cell organelles.

(4) Chloroplasts, endoplasmic reticulum and vacuole.

40. In *Amoeba*, osmoregulation is facilitated by

(1) Gas vacuole

(2) Phosphate granule

(3) Contractile vacuole

(4) Food vacuole

41. Which of the following cell organelle serves as the site of formation of glycoproteins and glycolipids?

(1) ER

(2) Vacuole

(3) Lysosome

(4) Golgi bodies

42. Nucleus as a cell organelle was first described by

(1) Flemming

(2) Robert Brown

(3) George Palade

(4) Camillo Golgi

43. Which of the given plastids store fats and oils?

(1) Elaioplast

(2) Amyloplast

(3) Aleuroplast

(4) Chromoplast

44. The infoldings of inner membrane of mitochondria are called

(1) Matrix

(2) Cisternae

(3) Cristae

(4) Pili

45. Thylakoids stack to form

(1) Stroma lamellae

(2) Grana

(3) Matrix

(4) Cristae

ZOOLOGY

46. The property that is found in muscle fibres and not in nerve fibres which enables muscle fibers to shorten its length, is

(1) Conductivity

(2) Contractility

(3) Excitability

(4) Insulation

47. Which of the given features is **not** related to cardiac muscle fibres?

(1) Involuntary and autorhythmic

(2) Branched and striated

(3) Cylindrical and have intercalated discs

(4) Structural syncytium and have communication junctions

48. Smooth muscles are **not** present in

(1) Wall of stomach

(2) Wall of intestine

(3) Wall of urinary bladder

(4) Biceps

49. Muscles fibres which taper at both ends are _______ .
Choose the option which fills the blank **correctly**

(1) Voluntary in action

(2) Cylindrical in shape

(3) Unbranched in nature

(4) Multinucleated in nature

50. The greatest control over responding to stimuli received by the body towards changing conditions is exhibited by

(1) Muscular tissue

(2) Epithelial tissue

(3) Neural tissue

(4) Adipose tissue

51. Neuroglial cells exhibit

(1) Cell division

(2) Excitability

(3) Conductivity

(4) Extensibility

52. In neural tissue the
A. Axon carries impulse towards the cyton
B. Axon carries impulse away from cyton.
C. Dendrite carry impulse away from cyton.
Select the **correct** option.

(1) A and B only

(2) A and C only

(3) Only A

(4) Only B

53. While analysing the chemical composition of the living tissue, the sample is generally grinded with the help of mortar and pestle in

(1) CH_3COOH

(2) Cl_3CCOOH

(3) NaCl

(4) $CH_3(CH_2)_{14}COOH$

54. The **incorrect** option w.r.t primary metabolites is that they

(1) Are required for basic metabolic processes

(2) Have identifiable functions

(3) Do not take part in physiological processes

(4) Include nucleic acids

55. The structures that contain polysaccharides are
a. Plant cell wall
b. Fungal cell wall
c. Exoskeleton of arthropods
Select the option with correct set.

(1) Only a

(2) Only b

(3) Only a and c

(4) All a, b and c

56. The component of cell that constitutes more % of the total cellular mass than that of proteins is

(1) Carbohydrates

(2) Lipids

(3) Water

(4) Ions

57. Which of the following is the monomeric unit of inulin?

(1) Glucose

(2) Fructose

(3) Starch

(4) Sucrose

58. Among the following secondary metabolites which is a toxin?

(1) Morphine

(2) Concanavalin A

(3) Vinblastine

(4) Ricin

59. All the molecules in the acid insoluble fraction are polymeric substances with the exception of

(1) Proteins

(2) Nucleic acids

(3) Polysaccharides

(4) Lipids

60. A polysaccharide which is utilised to form paper from pulp of plant and cotton fibre is

(1) Inulin

(2) Cellulose

(3) Starch

(4) Glucosamine